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Smaky 6 — User Guide

Smaky 6 rev 2-8, SAMOS OS, CLI.SY

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1. Overview

The **Smaky 6** is a Swiss microcomputer designed by Jean-Daniel Nicoud at EPFL (Lausanne) around 1978–1982. It is powered by a **Zilog Z80 at 2.5 MHz**, has **64 KB of RAM**, and a **text display of 20 lines × 64 characters** combined with a superimposable graphics layer of over 30,000 pixels.

The operating system is called **SAMOS** (*SMAky OS*). The user interface is handled by **CLISY**, a command-line interpreter loaded from the system disk at boot.

Hardware specifications:

Component	Detail
CPU	Zilog Z80, 2.5 MHz
RAM	64 KB
ROM (Phantom)	2 KB (boot manager)
Text display	20 lines × 64 columns
Graphics display	> 30,000 pixels, superimposable
Keyboard	QWERTZ Swiss layout, 57 keys + 7 function keys
Floppy	Micropolis, 77 tracks × 16 sectors × 256 bytes ≈ 315 KB/disk
Drives	DX0: (lower), DX1: (upper)
Serial interfaces	2 × USART (paper tape, modem)
Parallel interface	1 (printer, external device)
Speaker	Programmable buzzer (bit-banged)
Clock	Real-time clock with battery backup

2. Getting Started

Insert a **system disk** into drive **DX0:** (the lower drive) and power on the Smaky 6. The Phantom ROM displays a startup prompt.

Boot key combinations:

Combination	Effect
SHIFT - BREAK	Normal boot from DX0:
FUNCTION - SHIFT - BREAK	Boot from DX1:
BREAK	PDP-11 format loader (from USART 14)
FUNCTION - BREAK	Memory test (POST)

After loading, the > prompt appears — the system is ready.

Typical boot sequence:

```
ROM de chargement rev 1-7
SAMOS   rev 2-8
DX0:
>
```

3. The Command Line

The > prompt means the system is waiting for a command. Commands are **not** case-sensitive — LIST, list, and List are all equivalent.

Keyboard shortcuts:

Key	Effect
TAB	Inserts DX1: into the command line
BREAK (ESC)	Cancels current line; if empty, recalls last command
KILL (function key)	Aborts current peripheral transfer
SHIFT-BREAK	Hardware reset → reboot from DX0:

General syntax:

> COMMAND [argument]

Arguments containing spaces must be enclosed in single quotes (').

4. File Management

4.1 List Files

> LIST
> LIST DX1:
> LIST MYFILE.SM

Displays the contents of the current directory, or a specific drive/file. Columns show: name, size in bytes, protection attributes.

Equivalents: DOS DIR — Unix `ls -l`

4.2 Copy a File

> COPY SOURCE.SM
> COPY SOURCE.SM DEST.SM
> COPY DX0:SOURCE.SM DX1:DEST.SM

Copies a file on the same drive or from one drive to another. If the destination filename is not specified, the source name is kept.

Tip: Press TAB to insert DX1: automatically.

Equivalents: DOS COPY SOURCE DEST — Unix `cp SOURCE DEST`

4.3 Delete a File

> DELETE MYFILE.SM

Permanently deletes a file. No confirmation is requested. A file marked **permanent** (ERROR 4) cannot be deleted.

Equivalents: DOS DEL — Unix rm

4.4 Display File Contents

> TYPE MYFILE.BS

Displays the contents of a text file on screen. Press KILL to interrupt.

Equivalents: DOS TYPE — Unix cat / less

4.5 Print a File

> PRINT MYFILE.BS

> PRINT MYFILE.BS \$LP

Prints a file on the connected printer (requires LP.SY loaded).

Equivalents: DOS PRINT — Unix lpr

4.6 Append Files

> APPEND FILE1.BS FILE2.BS

Appends the contents of FILE2.BS to the end of FILE1.BS.

Equivalents: DOS COPY A+B A — Unix cat B >> A

4.7 Rename a File

> SET OLD.SM NEW.SM

Renames (or moves) a file. Can also modify protection attributes.

Equivalents: DOS REN — Unix mv

4.8 Compact Disk Space

> COMPRESS

Compacts the directory by removing deleted entries and consolidating free space. Run periodically to maintain performance.

Equivalents: DOS DEFRAG (partial analogy) — Unix: no direct equivalent.

5. Directories

The Smaky 6 uses **subdirectories** stored as files with the .DR extension.

```
> CDIR          - displays current directory
> CDIR PROJECTS.DR - enters the PROJECTS subdirectory
> CDIR ..       - goes up one level (if supported)
> CLEAR        - clears screen and returns to root directory
```

Equivalents: DOS CD / DIR — Unix cd / ls

6. Running Programs

To run a program (.SM file), simply type its name without the extension:

```
> SMILE          - launches the SMILE assembler
> PASCAL         - launches the Pascal interpreter
> BASIC          - launches the BASIC interpreter
```

.MC files are **CLI macros** (command scripts) and are invoked the same way. The system distinguishes them automatically.

Equivalents: DOS .EXE / .COM — Unix ./program

LOAD

```
> LOAD
```

Reloads the last executed program from disk. Useful after a crash or to quickly re-run the same program.

7. Floppy Disk Management

7.1 Initialize (Format) a Disk

```
> INIT DX1:
```

Formats and initialises a blank floppy in drive **DX1:**. **All data is erased.** The disk must be in the drive before running this command.

Equivalents: DOS FORMAT A: — Unix mkfs

7.2 Set Default Drive

> DX0:
> DX1:

Selects the default drive for subsequent operations. The TAB shortcut inserts DX1: directly in the command line.

Equivalents: DOS A: / B: — Unix cd /mnt/dx0

8. Display Modes

The Smaky 6 has two independent video layers: **text (alpha)** and **graphics**. The MODE command enables them separately or together.

> MODE — text-only mode (same as MODE A)
> MODE A — text only
> MODE G — graphics only
> MODE G P — graphics, fine-pixel variant
> MODE 2 — text AND graphics superimposed
> MODE 2 P — text + graphics, fine pixels

9. Date and Time

> STIME 15:30 — sets the time to 15:30
> SDATE 09/05/83 — sets the date to 9 May 1983
> SDAY SATURDAY — sets the day of the week

Equivalents: DOS TIME / DATE — Unix date -s "..."

10. Transfers and Peripherals

Peripherals are referenced with a \$ prefix in COPY, TYPE, PRINT, APPEND:

Name	Dir.	Hardware
\$PR	Input	Paper tape reader (USART 4, 20 mA loop)
\$PP	Output	Paper tape punch (USART 4)
\$PI	Input	Parallel interface
\$PO	Output	Parallel interface
\$MI	Input	Modem (USART 6)
\$MO	Output	Modem (USART 6)
\$LP	Output	Line printer (overlay LP.SY required)
\$KEY	Input	Keyboard
\$DIS	Output	Display

Examples:

```
> COPY MYFILE.BS $LP      - print to line printer
> COPY $PR MYFILE.BS      - read paper tape into a file
> COPY MYFILE.BS $M0      - send a file via modem
```

Press **KILL** to interrupt any transfer.

11. System Monitor

```
> MON
```

Enters the **machine monitor** (SYSMON). This low-level mode lets you examine and modify memory, execute machine code, and debug programs.

To return to the CLI command line, type the monitor's exit command.

Equivalents: DOS DEBUG — Unix gdb / xxd

12. Special Keys

Key	Context	Effect
TAB	Command line	Inserts DX1: into the line
BREAK / ESC	Command line	Cancels line; if empty, recalls previous one
KILL	Anywhere	Aborts current transfer or print job
SHIFT-BREAK	Boot / runtime	Hard reset → reboot from DX0:
FUNCTION-SHIFT-BREAK	Boot	Reboot from DX1:
FUNCTION-BREAK	Boot	Memory test (POST)
CHANGE (F1)	Programs	Program function key
SEARCH (F2)	Programs	Program function key
SHOW (F3)	Programs	Program function key
COPY (F4)	Programs	Program function key
CURSOR (F5)	Programs	Program function key
PROGRA (F6)	Programs	Program function key
KILL (F7)	Programs	Program function key / abort transfer

13. File Extensions

Extension	Type
.SY	System file (OS)
.SM	Executable program
.MC	CLI macro (command script)
.BS	BASIC source
.SR	SMILE assembler source
.DR	Directory (subdirectory)

Extension	Type
.LS	Assembler listing
.ST	Symbol table
.IM	1-bpp bitmap image
.HP	Help file (text)

14. Error Messages

Errors are displayed as ERROR nnn where nnn is the code in **octal**.

Dec	Oct	English message (manual)	French string (ER.SY)
1	001	Write protect file	fichier protégé écriture
2	002	Read protect file	fichier protégé lecture
4	004	Permanent file	fichier permanent
5	005	Line too long	ligne trop longue
6	006	End of file	fin de fichier
7	007	File end overflow	dépassement fin de fichier
10	012	File in use for writing	fichier ouvert en écriture
11	013	File already exists	fichier déjà existant
12	014	File does not exist	fichier inexistant
13	015	Illegal filename	nom de fichier illégal
14	016	Illegal reservation	réserve illégale
16	020	Cannot load file	chargement impossible
17	021	Out of file	plus de fichier
20	024	File in use for reading	fichier ouvert en lecture
21	025	Unknown device	périphérique inconnu
22	026	Channel error	erreur de canal
23	027	File(s) in use	fichier(s) en cours
24	030	All channels in use	tous les canaux occupés
25	031	Directory full	répertoire plein
26	032	Disk full	disque plein
30	036	Device timeout	timeout périphérique
31	037	Write protect tab set	languette de protection
32	040	Write error	erreur d'écriture
33	041	Read error	erreur de lecture
34	042	No starting address	pas d'adresse de départ
35	043	Bad load	chargement erroné
36	044	Buffer full	tampon plein
110	156	Illegal order	ordre illégal
114	162	System error	erreur système
115	163	Map error	erreur de map

Example: ERROR 043 = octal 043 = decimal 35 = *Bad load* (unreadable disk).

15. Quick Reference Table

Smaky 6 command	Description	DOS	Unix/Linux
LIST	List directory	DIR	ls -l
LIST DX1:	List drive DX1:	DIR B:	ls /mnt/dx1
CDIR NAME.DR	Change directory	CD NAME	cd NAME
COPY SRC DEST	Copy a file	COPY SRC DEST	cp SRC DEST
DELETE NAME	Delete a file	DEL NAME	rm NAME
SET OLD NEW	Rename a file	REN OLD NEW	mv OLD NEW
TYPE NAME	Display file contents	TYPE NAME	cat NAME
PRINT NAME	Print a file	PRINT NAME	lpr NAME
APPEND A B	Append files	COPY A+B A	cat B >> A
COMPRESS	Compact disk space	DEFRAG	(n/a)
INIT DX1:	Format a floppy disk	FORMAT B:	mkfs /dev/fd1
MODE A	Text-only display	MODE C080	(n/a)
MODE G	Graphics-only display	MODE BW320	(n/a)
MODE 2	Text + graphics	(n/a)	(n/a)
STIME HH:MM	Set the time	TIME	date -s "HH:MM"
SDATE DD/MM/YY	Set the date	DATE	date -s "..."
HELP	Online help	HELP	man / --help
MON	Machine monitor (debugger)	DEBUG	gdb / xxd
LOAD	Reload last program	(n/a)	(n/a)
STP	Stop / halt the system	(n/a)	halt
PROGNAME	Run a program	PROGNAME.EXE	./progname
DX0: / DX1:	Select default drive	A: / B:	cd /mnt/dx0
SHIFT-BREAK	Hard reset	Ctrl+Alt+Del	reboot

Document based on: original Smaky 6 rev 2-8 manual (EPFL/Epsitec), reverse-engineering of CLI.SY and SYS.SY, and the Smaky6emu emulator.